Arranging circles of radii $1, 2, \ldots, n$ around a central circle: a Supnick TSP and certified finite optima

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June 2026

Abstract

We study a discrete-geometric optimization problem: circles of radii $1, 2, \ldots, n$ are all externally tangent to a central circle, and the central radius R is minimized over cyclic orders of the surrounding circles. We prove that the chain-ordering component is governed by a fixed Supnick/anti-Monge traveling-salesman order. For every R, the angular-separation matrix is symmetric anti-Monge, so Supnick's theorem gives one minimizing cyclic order, independent of R. This proves the conjectured "pyramid" order optimal whenever the corresponding chain necklace is geometrically realizable, and gives an unconditional lower bound in all cases.

Full geometric feasibility can fail because non-adjacent circle constraints are not captured by the chain equation; from $n = 8$ the smallest circle can become a floating circle tangent only to the central circle. We formulate the full problem as a circular system of pairwise angular constraints, equivalently a simple temporal network, and certify global optima for $3 \le n \le 14$ using branch-and-bound plus an independent 50-digit verifier. We observe heuristically that the floating-circle cascade continues beyond the certified range, and we state the continuation and the asymptotic form $R^*(n) = n^2/8(1 + o(1))$ as conjectures. The repository contains the saved certificate artifacts, verifier, and reproducibility commands.

1 Introduction

The following question was asked by the user Dan on Mathematics Stack Exchange in January 2023 [1] and appears as question 295 in the UQ benchmark of unsolved Stack Exchange questions [2].

> Non-overlapping circles of radius $1, 2, 3, \ldots, n$ are all externally tangent to a middle circle. How should we arrange the surrounding circles in order to minimize the middle circle's radius R?

Dan conjectured a specific "pyramid-fashion" arrangement and observed two phenomena that make the problem subtle: (i) the natural closure equation

$$\sum_{k=1}^{n} \arccos\Big(1 - \frac{2 r_k r_{k+1}}{(R + r_k)(R + r_{k+1})}\Big) = 2\pi \qquad (r_{n+1} = r_1) \tag{1}$$

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treats consecutive circles as mutually tangent, yet (ii) for $n \geq 8$ the smallest circles can be "placed last anywhere they fit", i.e. the optimal configuration contains circles tangent to the central circle only. A comment by Rei Henigman [1] verified the conjecture for $n = 10$ by enumerating permutations of $\{2, \ldots, 10\}$, manually excluding circle 1 from the chain.

This paper addresses the problem on three levels, keeping the epistemic status of each result explicit.

1. The ordering problem is a Supnick TSP. Writing $\theta_R(a, b)$ for the minimum angular separation, at central radius R, between the centers of tangent circles of radii a and b (Section 2), we show that the matrix $\big(\theta_R(r_i, r_j)\big)_{i,j}$ is symmetric *anti-Monge* for every $R > 0$ (Lemma 5). Minimizing total angle over cyclic orders is therefore a *maximum* traveling salesman problem on a Supnick matrix, which by a classical theorem of Supnick [3] (see also the survey [4]) is solved by one fixed tour independent of the matrix entries:

$$\sigma^* = \langle 1,\ n-1,\ 3,\ n-3,\ 5,\ n-5,\ \ldots,\ n-2,\ 2,\ n \rangle.$$

A short self-consistency argument (the same order is optimal at *every* R) transfers optimality from fixed R to the variable-R problem (Theorem 6). The order σ^* coincides with Dan's conjectured pyramid arrangement. In particular $R_{\text{chain}}(\sigma^*)$, defined by (1), is an unconditional lower bound for $R^*(n)$, and it is the exact optimum whenever the σ^*-necklace is geometrically realizable.

2. Realizability breaks in a finite certified range. The closure equation (1) ignores non-adjacent pairs. For $n \geq 8$ the configuration it describes for σ^* is *not realizable*: circle 1, seated between circles n and $n-1$, no longer keeps them apart, i.e. $\theta_R(n, 1) + \theta_R(1, n-1) < \theta_R(n, n-1)$. Circle 1 then leaves the chain and becomes a *floating* circle, tangent to the central circle only. We model the full problem as a circular system of $\binom{n}{2}$ pairwise angular constraints — equivalently a simple temporal network (STN) — and give a certified algorithm for it (Section 4). Exhaustive certified optimization for $3 \leq n \leq 14$ (Section 5) reveals four regimes, governed by two closed-form quantities of the Supnick tour on $\{2, \ldots, n\}$: its geometric validity and the largest Descartes pocket it opens.

regime	n	floating set	optimal chain
I (full necklace)	$3 \leq n \leq 7$	$\varnothing$	σ^* on $\{1, \ldots, n\}$
II (paid floater)	$n = 8, 9$	$\{1\}$	distorted tour on $\{2, \ldots, n\}$
III (free floater)	$n = 10, 11, 12$	$\{1\}$	σ^* on $\{2, \ldots, n\}$
IV (second breakdown)	$n = 13$	$\{1\}$	constrained tour on $\{2, \ldots, n\}$
II$'$ (cascade, level 2)	$n = 14$	$\{1, 2\}$	constrained tour on $\{3, \ldots, n\}$

3. Quantitative results. We report certified optima, binding structures and floating sets for all $n \leq 14$, verified to 50 decimal digits (Table 1); the radius-maximizing (worst) arrangement, which is the *minimum*-tour counterpart $\langle 1, 3, 5, \ldots, n, \ldots, 4, 2 \rangle$ of Supnick's theorem (Proposition 8); and the conjectured asymptotics $R^*(n) \sim n^2/8$ (Section 6).

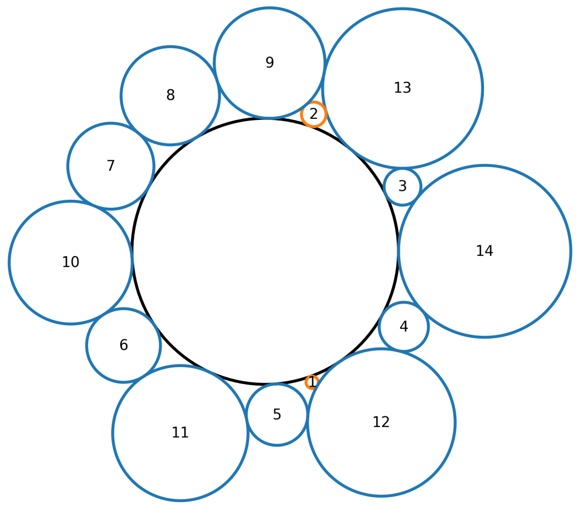

Figure 1: The certified optimum for $n = 14$; floating circles $1, 2$ are highlighted.

Epistemic ledger. Three levels of certainty are kept explicit throughout. *Proved*: Lemmas 1 and 5, Theorem 6, and the chain-level statement of Proposition 8. *Certified computation* (exhaustive search with explicitly stated numerical guards, independently re-checkable from the repository): Tables 1 and 3, the realizability statements of Propositions 9–11 on their stated finite ranges, and the three seam-failure computations. *Heuristic or conjectural*: Table 2, Conjecture 13, and Conjecture 14.

Related work. Supnick's fixed-tour theorem [3] and the Monge/Supnick taxonomy of polynomially solvable TSPs are surveyed in [4]. Circle packing literature concerns containers (packing circles *inside* regions, see e.g. [6]); the present problem — ordering unequal circles *around* a central one — appears not to have been treated. The tangency mechanics are governed by Descartes' Circle Theorem [5]. To our knowledge the link between Supnick matrices and rings of tangent circles is new.

2 The model

Let the central circle have radius $R > 0$ and center O. A surrounding circle of radius r tangent externally to it has its center at distance $R + r$ from O; its position is a single angle φ. The whole configuration is a vector of angles.

Lemma 1 (Exact angular reformulation)**.** *Circles of radii a, b, both externally tangent to the central circle, with angular positions separated by $\Delta \in [0, \pi]$, are non-overlapping if and only if*

$$\Delta \;\geq\; \theta_R(a, b) \;:=\; 2 \arcsin \sqrt{\frac{ab}{(R + a)(R + b)}}.$$

Moreover $\theta_R(a,b) \in (0,\pi)$ always, θ_R is symmetric, strictly decreasing in R, and strictly increasing in each radius.

Proof. By the law of cosines the squared distance between the centers is $(R+a)^2 + (R+b)^2 - 2(R+a)(R+b)\cos\Delta$, and non-overlap means this is at least $(a+b)^2$. Rearranging gives $\cos\Delta \le 1 - \frac{2ab}{(R+a)(R+b)}$, i.e. $\Delta \ge \theta_R(a,b)$ with θ_R as stated, using $\arccos(1-2s^2) = 2\arcsin s$ for $s \in [0,1]$. The argument of the square root is strictly less than 1 since $(R+a)(R+b) - ab = R(R+a+b) > 0$, whence $\theta_R < \pi$; the monotonicity claims are immediate from the formula. $\square$

Thus the problem is: minimize R such that there exist angles $\varphi_1, \dots, \varphi_n$ with circular separations $\ge \theta_R(r_i, r_j)$ for *all* pairs. For a fixed cyclic order this is a circular system of difference constraints — a simple temporal network — and feasibility at fixed R is decidable by shortest paths (Section 4). Two derived quantities recur:

Definition 2. For a cyclic order σ of a multiset of radii, $R_{\text{chain}}(\sigma)$ is the unique solution of $\sum_i \theta_R(\sigma_i, \sigma_{i+1}) = 2\pi$. (Existence and uniqueness: the left side tends to $|\sigma|\,\pi \ge 2\pi$ as $R \to 0^+$, to 0 as $R \to \infty$, and is strictly decreasing.) For a feasible pair (order, R), summing the adjacent gaps shows $\sum_i \theta_R(\sigma_i, \sigma_{i+1}) \le 2\pi$ and hence $R \ge R_{\text{chain}}(\sigma)$: the chain value is an unconditional per-order lower bound.

Definition 3 (Floating circle)**.** A circle c is *floating* at a global optimum if there exists an optimal configuration in which c is tangent only to the central circle (every pairwise constraint involving c is strictly slack).

Definition 4 (Descartes pocket)**.** For adjacent, mutually tangent circles a, b on the necklace at central radius R, the *pocket* is the largest circle externally tangent to the central circle and to both: by Descartes' Circle Theorem its curvature is $k_{\text{pocket}} = \frac{1}{R} + \frac{1}{a} + \frac{1}{b} + 2\sqrt{\frac{1}{Ra} + \frac{1}{ab} + \frac{1}{Rb}}$. A circle of radius ρ floats freely between a and b iff $\rho \le 1/k_{\text{pocket}}$, equivalently $\theta_R(a,\rho) + \theta_R(\rho,b) \le \theta_R(a,b)$.

3 The optimal cyclic order is a fixed Supnick tour

Lemma 5 (Supermodularity)**.** *For every $R > 0$ the matrix $\big(\theta_R(r_i, r_j)\big)$, rows and columns indexed by the radii in increasing order, is symmetric strictly anti-Monge: for $a < a'$ and $b < b'$,*

$$\theta_R(a,b) + \theta_R(a',b') \;>\; \theta_R(a,b') + \theta_R(a',b).$$

Proof. Write $\theta_R(a,b) = 2\arcsin(t_a t_b)$ with $t_x = \sqrt{x/(R+x)} \in (0,1)$, which is strictly increasing in x. For $f(x,y) = \arcsin(xy)$ on $(0,1)^2$,

$$\frac{\partial^2 f}{\partial x\,\partial y} = \frac{\partial}{\partial y}\,\frac{y}{\sqrt{1-x^2y^2}} = \frac{(1-x^2y^2) + x^2y^2}{(1-x^2y^2)^{3/2}} = (1-x^2y^2)^{-3/2} > 0,$$

so f is strictly supermodular; supermodularity is preserved by strictly increasing reparametrizations of each coordinate, and the claim follows. $\square$

Equivalently, $-\theta_R$ is a Supnick matrix — symmetric Monge off the diagonal, diagonal entries being irrelevant for tours (cf. [4, Prop. 2.13]). Supnick's theorem [3] covers both extreme tours; in the form stated in [4] (Theorem 2.5 for the minimum, and the max-TSP discussion following

Theorem 2.12, both attributed there to [3]), the minimum TSP tour on Supnick matrices is $\langle 1, 3, 5, \ldots, n, \ldots, 6, 4, 2\rangle$ and the *maximum* TSP tour is

$$\sigma^* = \langle 1,\ n-1,\ 3,\ n-3,\ 5,\ n-5,\ \ldots,\ n-2,\ 2,\ n\rangle,$$

independently of the matrix entries. Maximizing $\sum(-\theta_R)$ is minimizing $\sum \theta_R$, hence:

Theorem 6 (Optimal order)**.** *Let $r_1 < \cdots < r_n$ be arbitrary positive radii.*

(i) For every $R > 0$, among all cyclic orders the tour σ^* minimizes $\sum_i \theta_R(r_{\sigma(i)}, r_{\sigma(i+1)})$.

(ii) σ^* minimizes R_{chain} over all cyclic orders; hence $R^* \geq R_{\mathrm{chain}}(\sigma^*)$ unconditionally.

(iii) If the σ^*-necklace at $R_{\mathrm{chain}}(\sigma^*)$ is geometrically realizable (no non-adjacent pair violated), then it is a global optimum: $R^* = R_{\mathrm{chain}}(\sigma^*)$.

Proof. (i) is Lemma 5 plus Supnick's theorem. (ii): fix any order σ and let $R = R_{\mathrm{chain}}(\sigma)$. By (i), $\sum \theta_R(\sigma^*) \leq \sum \theta_R(\sigma) = 2\pi$; since $\sum \theta_R(\sigma^*)$ is strictly decreasing in R and equals 2π at $R_{\mathrm{chain}}(\sigma^*)$, we get $R_{\mathrm{chain}}(\sigma^*) \leq R$. The unconditional bound follows since every feasible (σ, R) has $R \geq R_{\mathrm{chain}}(\sigma)$. (iii): realizability makes the lower bound attained. □

Remark 7. Theorem 6 holds for *arbitrary* distinct radii, not only $1, \ldots, n$; the arithmetic sequence enters only in the realizability analysis below. For $n = 10$ the lower bound of (ii) is ≈ 9.90, the value observed in [1] when circle 1 is (incorrectly) kept in the chain; the true optimum is $9.9799\ldots$.

Proposition 8 (Worst arrangement)**.** *For every R the cyclic order maximizing $\sum_i \theta_R$ is the complementary Supnick tour $\sigma_\dagger = \langle 1, 3, 5, \ldots, n, \ldots, 6, 4, 2\rangle$; consequently $\sigma_\dagger$ maximizes R_{chain}. For all $3 \leq n \leq 13$ the $\sigma_\dagger$-necklace is realizable, so among necklace arrangements it maximizes the central radius exactly; certified values in Table 3.*

4 Exact algorithm and certification

Feasibility oracle. Fix a cyclic order and R. Unrolling the circle at one circle ($\varphi_0 = 0$) gives, for every pair $i < j$ of positions, the two constraints $\varphi_j - \varphi_i \geq \theta_R(r_i, r_j)$ and $\varphi_j - \varphi_i \leq 2\pi - \theta_R(r_i, r_j)$. This is a simple temporal network; feasibility is the absence of a negative cycle in the associated distance graph and is decided by Floyd–Warshall in $O(n^3)$. The minimum feasible R for the order, $R_{\mathrm{full}}(\sigma)$, is found by bisection (the constraint set relaxes monotonically in R), bracketed below by $R_{\mathrm{chain}}(\sigma)$. A witness configuration is recovered from shortest-path potentials and verified independently in Cartesian coordinates.

Lower bounds and search. For the search over orders we use

$$\mathrm{LB}(\sigma) = \max\left\{R_{\mathrm{chain}}(\sigma),\ R_{\mathrm{chain}}(\sigma \setminus \{1\}),\ R_{\mathrm{chain}}(\sigma \setminus \{1,2\})\right\},$$

where $\sigma \setminus S$ denotes the induced cyclic order (valid since induced adjacencies are a subset of the full pairwise constraints). Orders are enumerated canonically (largest radius pinned, reflections halved), LB is computed vectorized for all $(n-1)!/2$ orders, and R_{full} is evaluated in ascending LB until the next bound meets the incumbent: this certifies global optimality up to an absolute guard of 10^{-10} in R. The guard dominates the floating-point error budget: bisections run to relative tolerance 10^{-13}; one bound evaluation is an n-term sum of arcsin values, with relative error $O(n\,\varepsilon_{64})$; and the float64 bounds were calibrated against 50-digit recomputation on random samples of orders, with maximal observed deviation 1.8×10^{-14} (over 10^5 random orders for each $8 \le n \le 14$; maximal overestimate 1.5×10^{-14}), four orders of magnitude below the guard. Independently of the search code, the repository ships a standalone verifier (`verify.py`) that re-derives each claimed optimum and re-checks every order on the pruning frontier (between 1 and 11 orders per n) in 50-digit arithmetic. Optimal binding structures were re-verified in 50-digit arithmetic (mpmath); the values in Table 1 are exact to all displayed digits, with maximal residual slack below 10^{-40}. The $n = 13$ certification enumerates $12!/2 \approx 2.4 \times 10^8$ orders (≈ 1.8 h on a consumer 8-core laptop); $n = 14$ enumerates $13!/2 \approx 3.1 \times 10^9$ orders (≈ 22 h). Code and artifacts: `https://github.com/falker47/ringmin`.

Essential binding structure. A pairwise constraint is *essential* if perturbing it alone by $+\varepsilon$ makes the STN infeasible at R^*; a circle is floating iff it carries no essential constraint. This distinguishes true tangencies from placement artifacts of a particular witness.

5 Certified optima for $n \le 14$ and the regime cascade

Table 1: Certified optima, rounded to 12 decimals from the 50-digit verified values (30-digit table in the Appendix). Cycles list radii in circular order; the floating circle is shown in its (non-unique) hosting gap. "Essential pair" is the tangent non-adjacent pair bridged by the floater.

n	$R^*(n)$	optimal cycle	floating	essential pair
3	0.260869565217 (= 6/23)	(3,1,2)	—	—
4	0.844453589561	(4,1,3,2)	—	—
5	1.695494081203	(5,1,4,3,2)	—	—
6	2.794919518897	(6,1,5,3,4,2)	—	—
7	4.153189553744	(7,1,6,3,4,5,2)	—	—
8	5.767794284590	(8,1,6,4,5,3,7,2)	$\{1\}$	$(8, 6)$
9	7.726726552611	(9,2,8,1,5,6,4,7,3)	$\{1\}$	$(8, 5)$
10	9.979907385863	(10,2,9,4,7,1,6,5,8,3)	$\{1\}$	$(7, 6)$
11	12.488720487188	(11,2,10,4,8,6,7,1,5,9,3)	$\{1\}$	$(7, 5)$
12	15.258870430448	(12,2,11,4,9,6,7,8,5,1,10,3)	$\{1\}$	$(5, 10)$
13	18.317563047217	(13,3,1,12,2,10,6,8,7,9,5,11,4)	$\{1\}$	$(3, 12)$
14	21.665395182215	(14,3,13,2,9,8,7,10,6,11,5,1,12,4)	$\{1, 2\}$	$(13, 9)$, $(5, 12)$

For $n \le 7$ the optimal cycle is exactly σ^* on $\{1, \dots, n\}$ (regime I), as guaranteed by Theorem 6(iii) once realizability is checked. The transitions are governed by two computable properties of the Supnick tour on the reduced set $\{2, \dots, n\}$:

Proposition 9 (First breakdown)**.** *In the σ^*-necklace on $\{1, \dots, n\}$, circle 1 sits between n and $n-1$. The realizability inequality $\theta_R(n, 1) + \theta_R(1, n-1) \ge \theta_R(n, n-1)$ at $R = R_{\text{chain}}(\sigma^*)$ holds for $n \le 7$ and fails for $8 \le n \le 13$ (violated pair $(n, n-1)$; at $n = 13$ also $(13, 11)$ across circle 2). Hence regime I ends at $n = 7$. We conjecture failure for all $n \ge 8$; cf. open problem (1).*

Proposition 10 (Free vs. paid floater)**.** *Let $\rho_{\max}(n)$ be the largest Descartes pocket of the σ^*-necklace on $\{2, \dots, n\}$ at its chain radius. Then* $\rho_{\max} = 0.858, 0.988, 1.139, 1.262, 1.409$

for $n = 8, \dots, 12$. For $n = 10, 11, 12$ one has $\rho_{\max} > 1$: circle 1 floats freely and $R^*(n) = R_{\text{chain}}\big(\sigma^*$ on $\{2, \dots, n\}\big)$ exactly (regime III). For $n = 8, 9$, $\rho_{\max} < 1$: the optimum distorts the reduced tour to open a unit pocket, at certified extra cost $\delta(8) = 0.04080$, $\delta(9) = 0.00532$ over the (then unattainable) reduced chain bound (regime II).

Proposition 11 (Second breakdown)**.** *At $n = 13$ the Supnick tour on $\{2, \dots, 13\}$ is itself unrealizable: circle 2, seated between 13 and 12, violates $\theta_R(13, 2) + \theta_R(2, 12) = 0.80448 < 0.83488 = \theta_R(13, 12)$ at $R = 18.3176$. The certified optimum demotes circle 2 to the pair $(12, 10)$, where the separation constraint holds with slack 6×10^{-3} rad; circle 1 floats in the $(3, 12)$ pocket (radius 1.086). Thus the mechanism that expelled circle 1 at $n = 8$ reaches circle 2 at $n = 13$ (regime IV).*

Remark 12 (Structural summary). For all $n \leq 14$ the optimum has the form: a single tangent necklace carrying a Supnick-like order *constrained to geometric realizability*, plus floating circles in Descartes pockets. The cascade continues self-similarly: circle 2 first floats at $n = 14$ (now *certified*, Table 1) via a *paid* distortion that opens a pocket of radius 2.013 between circles 13 and 9 — the regime-II mechanism one level up, at extra cost 0.0565 over the (then unattainable) chain bound of the Supnick tour on $\{3, \dots, 14\}$. At $n = 16$ the best known value coincides to 10^{-10} with the chain value of the Supnick tour on $\{3, \dots, 16\}$, the level-2 analogue of regime III. Heuristically, circle 3 joins by $n = 18$ (Table 2). Moreover the Supnick tour on $\{3, \dots, n\}$ itself becomes unrealizable at $n = 17$ — circle 3 failing at the $(17, 16)$ seam with slack -0.0104 — so the *seam-failure* onsets for circles $1, 2, 3$ are $n = 8, 13, 17$: each *computed* cascade level reproduces the full regime pattern of the previous one. The float onsets $n = 8, 14, 18$ for circles $1, 2, 3$ are consistent with Dan's $n/4$ Descartes-based upper bound on the number of ignorable circles [1]. Whether the cascade continues indefinitely is the content of the following conjecture.

Conjecture 13 (Cascade)**.** *For every $k \geq 1$ there is a threshold s_k such that the Supnick tour on $\{k, \dots, n\}$ is geometrically unrealizable for all $n \geq s_k$ (observed and certified: $s_1 = 8$, $s_2 = 13$, $s_3 = 17$), and circle k is floating in every optimal configuration for all sufficiently large n, with each level exhibiting the paid-then-free regime pattern of Propositions 9–11.*

Table 2: Best known arrangements for $15 \leq n \leq 18$ (local search, 200–600 restarts per n; *non-exhaustive*). The last column counts restarts reaching the best value.

n	best R	floating	cycle	hits
15	25.290151636714	$\{1, 2\}$	(15,3,14,5,12,2,9,1,8,10,7,11,6,13,4)	59
16	29.194988498141	$\{1, 2\}$	(16,1,4,14,6,12,8,10,9,2,11,7,13,5,15,3)	18
17	33.394425022062	$\{1, 2\}$	(17,4,16,3,15,6,1,13,8,11,10,9,2,12,7,14,5)	19
18	37.870755667393	$\{1, 2, 3\}$	(18,4,17,3,14,8,12,10,11,9,2,13,7,15,6,16,1,5)	9

6 Asymptotics

For $R \gg n$ one has $\theta_R(a, b) = 2\sqrt{ab}/R\big(1 + O(n/R)\big)$, so the closure equation gives $R \approx \frac{1}{\pi}\sum_i \sqrt{r_i r_{i+1}}$. Since $\sqrt{ab}$ is supermodular, Theorem 6(i) applies verbatim: the minimizing order of $\sum \sqrt{r_i r_{i+1}}$ is again σ^*, whose value on $1, \dots, n$ is

$$\min_{\sigma} \sum_i \sqrt{r_{\sigma(i)} r_{\sigma(i+1)}} = \Big(1 + o(1)\Big) \int_0^n \sqrt{x(n - x)}\, dx = \frac{\pi n^2}{8}\,(1 + o(1)),$$

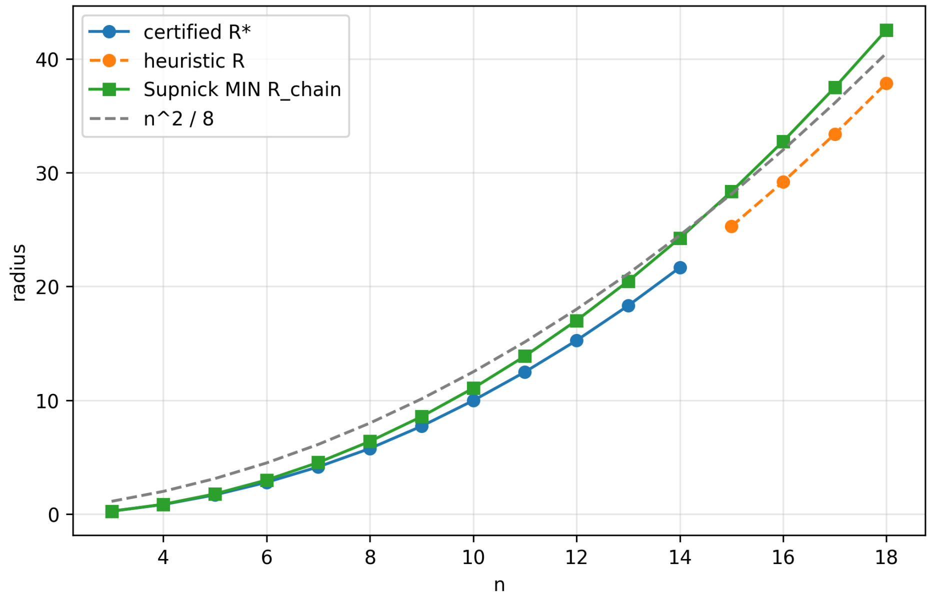

Figure 2: $R^*(n)$, the worst-arrangement curve, and the $n^2/8$ asymptote.

because adjacent pairs in σ^* are $\{k,\, n-k+O(1)\}$ with each odd k contributing twice. Self-consistency ($R \sim n^2/8 \gg n$) further requires the floating circles to be asymptotically negligible, in the sense $\sum_{k\in F(n)} \sqrt{kn} = o(n^2)$ — guaranteed if $|F(n)| = o(n)$, consistent with $|F(18)| = 3$, but open (problem (2) below). Under this assumption we are led to:

Conjecture 14. $R^*(n) = \dfrac{n^2}{8}\big(1+o(1)\big)$. *Empirically the deficit* $n^2/8 - R(n)$ *equals* 2.52, 2.64, 2.74, 2.81, 2.83 *for the certified* $n = 10, \ldots, 14$, *and* 2.83, 2.81, 2.73, 2.63 *for the heuristic* $n = 15, \ldots, 18$ *(the latter are* lower *bounds on the true deficit, heuristic radii being upper bounds for* R^**): bounded so far, and in any case growing strictly slower than* n. *We tentatively conjecture the stronger form* $n^2/8 - R^*(n) = O(\sqrt{n}\,)$.

Two-sided bounds making the leading order rigorous (via $\arcsin x \geq x$ and the explicit σ^* configuration as an upper bound) appear within reach but are not pursued here.

7 Open problems

(1) Prove Propositions 9–11 for all n analytically (the inequalities are explicit; monotonicity in n is the missing step). (2) Characterize the floating set $F(n)$ asymptotically (float onsets: circle 1 at $n = 8$ and circle 2 at $n = 14$, both certified, circle 3 at $n = 18$ heuristically; certified seam-failure onsets $n = 8, 13, 17$); is $|F(n)| \sim cn^{1/2}$, or does Dan's $n/4$ bound give the right order? Both are subsumed by proving Conjecture 13. (3) The same machinery applies to radii $r_k = k^\alpha$ or arbitrary sequences: Theorem 6 is general, the cascade analysis is not. (4) Three dimensions: spheres of radii $1, \ldots, n$ tangent to a central sphere — the ordering structure disappears; what replaces Supnick?

Acknowledgments

The problem and the pyramid conjecture are due to Dan [1]; Rei Henigman's $n = 10$ computation provided an independent check. The project used AI assistance for software, experimentation, drafting, and repository organization under the author's direction and final review. The paper is written to be independently checkable: the proof of Theorem 6 is short and self-contained, the only imported result is Supnick's theorem, and every certified value is reproducible from the public repository with the independent verifier described in Section 4.

References

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A High-precision values and figures

Table 3: The radius-maximizing necklace arrangement $\sigma_\dagger$ (Proposition 8): certified chain values, all geometrically realizable.

n	3	4	5	6	7	8	9	10	11	12	13
$R_\dagger$	0.2609	0.8673	1.7786	3.0001	4.5375	6.3934	8.5690	11.0647	13.8806	17.0164	20.4720

Per-n configuration figures for all $3 \le n \le 18$ are available in the repository.

n	R^*
3	0.260869565217391304347826086957
4	0.844453589560855604347528524674
5	1.69549408120271081351328017371
6	2.79491951889692485617024406797
7	4.15318955374381246513863858202
8	5.7677942845896143026361805725
9	7.72672655261128921886246177604
10	9.97990738586347760966552641468
11	12.4887204871876588517468786264
12	15.2588704304484933617250043503
13	18.3175630472173206282821941532
14	21.6653951822145150956462891793

Table 4: Certified optima, 30 significant digits.

n	Supnick min tour	R_{chain}	R_{full}
3	[1, 3, 2]	0.260869565217	0.260869565217
4	[1, 3, 4, 2]	0.867281640255	0.867281640255
5	[1, 3, 5, 4, 2]	1.778550949884	1.778550949884
6	[1, 3, 5, 6, 4, 2]	3.000138442323	3.000138442323
7	[1, 3, 5, 7, 6, 4, 2]	4.537467833562	4.537467833562
8	[1, 3, 5, 7, 8, 6, 4, 2]	6.393358320114	6.393358320114
9	[1, 3, 5, 7, 9, 8, 6, 4, 2]	8.568988935955	8.568988935955
10	[1, 3, 5, 7, 9, 10, 8, 6, 4, 2]	11.064726494858	11.064726494858
11	[1, 3, 5, 7, 9, 11, 10, 8, 6, 4, 2]	13.880577932626	13.880577932626
12	[1, 3, 5, 7, 9, 11, 12, 10, 8, 6, 4, 2]	17.016408238128	17.016408238128
13	[1, 3, 5, 7, 9, 11, 13, 12, 10, 8, 6, 4, 2]	20.472039120681	20.472039120681
14	[1, 3, 5, 7, 9, 11, 13, 14, 12, 10, 8, 6, 4, 2]	24.247291190092	24.247291190092
15	[1, 3, 5, 7, 9, 11, 13, 15, 14, 12, 10, 8, 6, 4, 2]	28.342000288518	28.342000288518
16	[1, 3, 5, 7, 9, 11, 13, 15, 16, 14, 12, 10, 8, 6, 4, 2]	32.756022334220	32.756022334220
17	[1, 3, 5, 7, 9, 11, 13, 15, 17, 16, 14, 12, 10, 8, 6, 4, 2]	37.489233291550	37.489233291550
18	[1, 3, 5, 7, 9, 11, 13, 15, 17, 18, 16, 14, 12, 10, 8, 6, 4, 2]	42.541527273519	42.541527273519

Table 5: Worst-arrangement comparison; Supnick min tour values.